CHAPTER 8

INTERSECTIONS OF SOLIDS

8.1. Polyhedra intersections

To intersect two polyhedra means to find the points the edges of one polyhedron intersect the faces of the other polyhedron on and reciprocally. Joining together these points gives the intersection polygons, that can be plan or spatial.

If. the result of the intersection is a single polygon, the intersection is type breaking, and if from intersection result two polygons, the intersection is type penetration.

In order to find the vertices of the intersection polygon, auxiliary planes are used, as follows:

- for the intersection of two prisms, the auxiliary planes are drawn along the edges, respectively through the vertices of the basis polygon;
- for the intersection prism – pyramid, the auxiliary planes are drawn through the vertex of the pyramid, are parallel to the direction of prism's edges, respectively along the line parallel to the prism's edges and through the vertex of the pyramid;
- for the intersection of two pyramids, the auxiliary planes are drawn through the edges and the line joining together the vertices of the two pyramids.

In all cases, some of the auxiliary planes are useful, the ones drawn through the edge of a polyhedron that intersect the basis of the other polyhedron. The two auxiliary planes framing the rest of the useful planes

are named boundary planes. The boundary planes also limit the parts of polyhedron (the edges) not involved in intersection. As these planes intersect the bases of the polyhedra, the type of intersection may be settled after drawing the planes.

In order to find the intersection polygon of two polyhedra the following stages are advised:

1. finding the useful auxiliary planes and the boundary planes;
2. finding the intersection points of the edges and the faces of the polygon;
3. joining together the intersection points previously found;
4. studying the visibility of the section polygon.

8.1.1. Intersection prism – prism

Given the inclined triangular prisms having the bases *ABC* and *MNQ* situated in the horizontal plane *[H]*. Find the intersection polygon of the two polyhedra (fig. 8.1).

Through an arbitrarily chosen point *S* (*s*, *s'*), the lines *(D)* and *(Δ)*, parallel to the edges of the prism, are drawn. The horizontal traces, *H* and H_1, of the two lines are found. These points settle the direction of the horizontal traces of the boundary and auxiliary planes, denoted *P*.

Fig. 8.1

Parallel to this direction, following the vertices of the two basis polygons of the two polyhedra, the auxiliary planes $[P_1]$, $[P_2]$, $[P_3]$ are drawn. Planes $[P_1]$ and $[P_3]$ are boundary planes, indicating that from the bases of the two polyhedra a part of each basis remained out of the intersection (hatched zone), respectively the edges adequate to the vertices *C* and *Q*. The intersection is type breaking.

Plane $[P_1]$ drawn through the vertex *A* intersects the basis *MNQ* in the points of projections (*1*, *1'*) and (*2*, *2'*).

Lines parallel to the edges of the prism with the basis *MNQ* drawn through points *1* and *2* intersect the edge adequate to the vertex *a* in the points a_1 and a_2. These are the points the edge adequate to the vertex *a* intersects the prism with the basis *MNQ* in.

The auxiliary plane $[P_2]$ drawn through the vertex *N* intersects the basis *ABC* in the points of projections (*3*, *3'*) and (*4*, *4'*). The intersection

points of the edge adequate to the vertex n with the prism having the basis ABC are n_3 and n_4.

Plane $[P_3]$ drawn through the vertex M intersects the basis ABC in the point B (b, b') and in the point of projections (5, $5'$). The intersection points of the edge adequate to the vertex m with the prism having the basis ABC are m_5 and m_b.

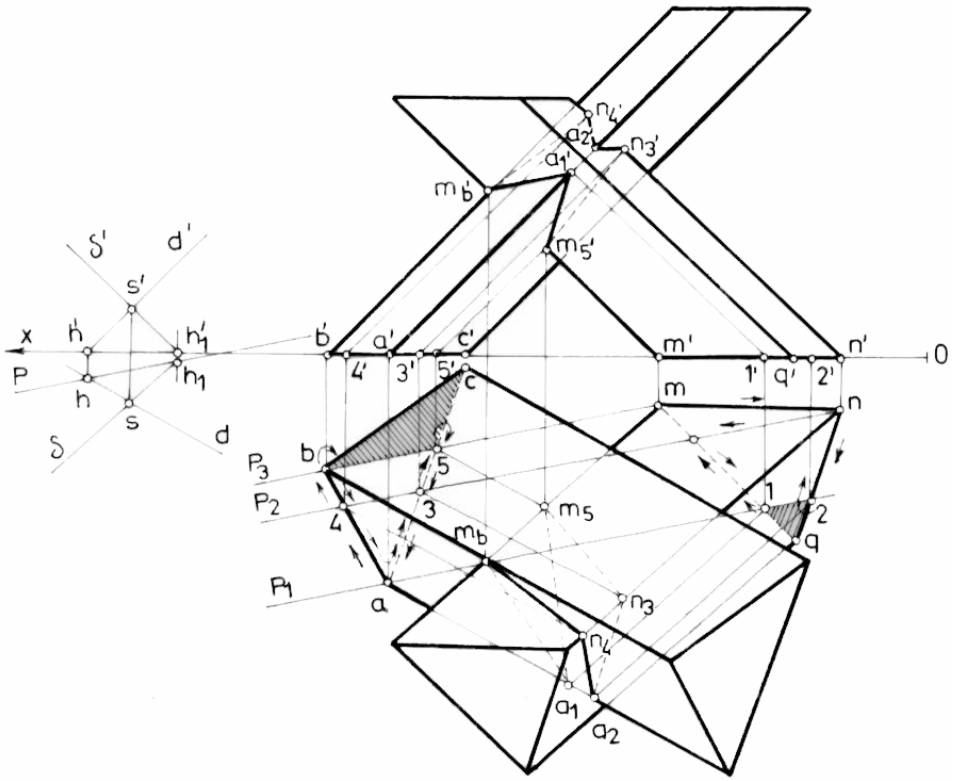


Figure 8.1

The vertical projections of the intersection points previously found are obtained with the help of the lines of recall drawn to the adequate edges. The order of joining together the vertices of the intersection polygon is settled by the mobile point method.